Advanced Topics in Deep Learning

2nd Semester - 2025/2026

Project Assignment – Advanced Time Series Forecasting, Optimization and Explainability

Version 1.0 (2026-02-01)

- **Due date:** April 5th at 23:59 (no late submissions)
- **Group work:** this assignment must be performed in groups of three students.
- **Language:** your project may be written either in Portuguese or in English (preferred).

1. Introduction

In the Time Series Modelling mini-project, you developed an end-to-end deep learning pipeline for multi-step forecasting, focusing on correct experimental design, model comparison, and understanding how different modelling choices influence forecasting behaviour.

This Final Project should be seen as a natural evolution of that work.

The objective now is to move beyond exploratory modelling and manual experimentation, and to design a systematic, optimized and well-analysed forecasting system, integrating advanced techniques covered throughout the course, namely:

- Evolutionary Algorithms for automated optimization,
- Advanced data preparation and windowing strategies,
- Generative approaches for data augmentation,
- Explainable AI (XAI) for model interpretation and trust,
- Efficiency and resource-aware analysis of deep learning models.

In the mini-project, performance was secondary to interpretation and learning.

In this Final Project, performance becomes a primary goal, but it must be achieved through systematic optimization and supported by rigorous analysis, interpretability, and efficiency trade-offs.

2. Dataset: Jena Climate (Weather)

This project builds upon the same dataset used in the Time Series Modelling mini-project, allowing you to leverage prior knowledge while substantially extending the scope and depth of the analysis.

2.1 Dataset Description

The Jena Climate dataset contains meteorological measurements collected at a weather station in Jena, Germany, between 2009 and 2016.

The dataset is publicly available and can be accessed at: https://keras.io/examples/timeseries/timeseries_weather_forecasting

Key characteristics:

- Original sampling frequency: every 10 minutes
- Number of variables: (originally) 15 meteorological features
- Time span: several years of continuous measurements
- Data type: real-valued, multivariate time series

Each timestamp corresponds to a single observation, and each column corresponds to a physical meteorological variable. Examples of variables include:

- air temperature,
- atmospheric pressure,
- relative humidity,
- wind speed and direction.

2.2 Target Variable

The target variable for this project is: **Air temperature (T (degC))**

Your task is to predict the future air temperature over a given time horizon, using the historical values of all available meteorological variables.

2.3 Variables to use

To ensure a balanced and non-trivial forecasting problem, the experiments in this project must use the following **input variables**:

- Date Time - Date-time reference
- T (degC) — past air temperature values
- p (mbar) — atmospheric pressure
- rh (%) — relative humidity
- wv (m/s) — wind speed
- max. wv (m/s) — maximum wind speed
- wd (deg) — wind direction

All other variables must not be used in your experiments.

3. Forecasting Task Definition

3.1 Problem Formulation

The forecasting problem must be formulated as **supervised multi-step forecasting**, as follows:

$$\mathbf{X}_{t-L+1:t} \longrightarrow \mathbf{y}_{t+1:t+H}$$

where:

- L is the **lookback window length** (number of past time steps used as input),
- H is the **forecast horizon** (number of future time steps to predict),
- $\mathbf{X}$ contains the meteorological variables,
- $\mathbf{y}$ contains only the future *air temperature*.

In other words, you will solve a:

Multivariate-input, univariate-output, multi-step forecasting problem

3.2 Suggested Base Configuration

You may start from the best configuration you achieved in the Time Series Modelling mini-project..

However, in this project, windowing choices are not fixed.

Different window configurations must be explored and optimized, potentially including:

- different lookback lengths,
- different forecast horizons,
- multi-scale or alternative windowing strategies.

4. Model Inputs

4.1 Mandatory Inputs

Model inputs must include:

- historical values of the meteorological variables, including past temperature values.

This ensures a genuinely multivariate forecasting problem.

4.2 Temporal Covariates

Temporal covariates may be included in the model inputs.

You may either:

- reuse the same temporal covariates explored in the Time Series Modelling mini-project, or
- propose and experiment with new temporal representations or encodings.

In all cases, any temporal covariate included in the model must be:

- clearly defined,
- appropriately encoded (e.g., preserving cyclic structure when relevant),
- and explicitly justified in the report.

The selection, inclusion, and representation of temporal covariates can be considered an integral part of the optimization and analysis process, and can impact on model performance, interpretability, and efficiency.

5. Tasks

Your project must follow and extend the "typical" Time Series forecasting pipeline.

5.1 Baseline Deep Learning Models

You must select at least one of the base architectures used in the mini-project (GRU or Transformer) and treat it as a baseline.

You are expected to use it as a reference for all subsequent improvements.

5.2 Evolutionary Optimization

You must apply Evolutionary Algorithms to automatically optimize the end-to-end forecasting pipeline, rather than isolated model hyperparameters. The optimization process may include, but is not limited to, the following components:

- model hyperparameters,
- architectural choices,
- windowing strategies,
- data preprocessing techniques,
- Other techniques that improve solution performance,
- combinations of the above.

You must clearly define:

- the genotype representation (what is being optimized and how it is encoded),
- the fitness function(s) used for evaluation, (the evolutionary fitness must be computed on a validation set only)
- the evolutionary strategy and operators (e.g. selection, mutation, crossover),
- the computational budget (population size, number of generations, training budget per individual).

Multi-objective optimization (e.g. error vs. model complexity or efficiency) is encouraged.

Note: Since evolutionary search may overfit to the validation set, you should adopt mitigation strategies (e.g., reevaluating the best individuals with multiple random seeds, keeping an untouched final test set, and reporting performance variability).

5.3 Synthetic Data Generation (Optional)

Optional (Bonus): You may explore generative approaches (e.g. GAN-based models) to generate synthetic time series segments, with the goal of:

- data augmentation,
- robustness improvement,
- or better generalization.

This component may strengthen the performance of the developed solution.

Important note on data leakage: If synthetic data generation is used, it is essential to avoid any form of data leakage. Synthetic data must be generated using the training set only, and its impact must be evaluated by comparing the final models with and without synthetic augmentation.

5.4 Explainable AI

You must apply Explainable AI (XAI) techniques to analyse your models.

This must include:

- at least one global explanation method,
- at least one local explanation method.

5.5 Efficiency and Resource Analysis

A key component of this project is the analysis of efficiency and computational cost, including:

- training time,
- inference time,
- number of model parameters,
- memory usage (RAM and/or GPU),
- Other items which you considered relevant.

You should analyse trade-offs between performance and efficiency.

5.6 Comparative Analysis and Discussion

This is one of the most important parts of the project.

Your discussion must include:

- comparison between baseline and optimized models,
- impact of evolutionary optimization,
- insights from explainability analyses,
- trade-offs between accuracy, complexity and efficiency,

- strengths and limitations of the final solution.

Unlike the mini-project, achieving the best possible model is an explicit objective, but it must be supported by solid analysis and justification.

6. Deliverables

Your submission should include:

- A Jupyter Notebook with your code and experiments.
    - The submitted notebook must include all executed outputs, so that the workflow and results are fully visible without requiring re-execution.
    - Additionally, you must ensure that the notebook can be easily executed in a new environment.
- Written report including:
    - introduction,
    - methodology,
    - experiments,
    - results,
    - discussion,
    - conclusion.

Note: Writing this report does not replace the need for commenting on your code. You should add inline explanations, observations and reflections directly in the notebook, so that your reasoning process is clear during experiments.

If you prefer, you can also deliver your analysis (short report) in the notebook. It should be presented in text form, possibly with supporting images.

7. Evaluation

Your work will be evaluated based on:

- Ability to understand and address the challenge,
- Correctness and completeness of the forecasting pipeline,
- Completeness and rigor of the experiments performed,
- Correctness and soundness of the evolutionary optimization design (search space, fitness definition, fairness, and budget discipline),
- Depth and relevance of the explainability (XAI) analysis,
- Quality of efficiency and resource evaluation,
- Clarity and depth of the discussion and conclusions,
- Organization, correctness and reproducibility of the code.

8. Requirements and Submission Guidelines

Please note the following requirements for your submission:

- The project must be carried out in groups of 3 students.

- Submission deadline: 5-april at 23:59.
- The project report may be written either in Portuguese or in English (English is preferred).
- Your notebook and report must be well organized and follow the good practices, including:
  - Use of clear sections and structure.
  - Integration of code with explanations and commentary.
  - Compliance with coding conventions and readability standards.
  - Ensuring reproducibility of experiments.
  - Proper handling of imports and installation of dependencies (if required).
- Ensuring reproducibility is always a good idea: fixed seeds, library requirements, and clear execution instruction.
- The submitted notebook must include all executed outputs, so that the workflow and results are fully visible without requiring re-execution.
- You must ensure that the notebook can be easily executed in a new environment, including the professor environment. This implies:
  - Providing clear installation and execution instructions,
  - Including any configuration necessary to reproduce the results.
- Submit through Moodle: notebook, report and any other required files

9. Remarks

- You can copy code from the web, but
  - you must always provide the reference of the code that you copy
  - Make sure that you explain any external code so that another student in the course understands it and can reuse it.
- During the development stage of your work, you should perform as many tests as you wish. However, you must submit a clean version of your notebook, containing only the relevant steps. In order to do so, before submitting you should copy your development notebook into a new file and clean it by deleting spurious tests and code.

- It is not always necessary to have a full-blown analysis, but you are expected to think! For each step, ask yourself why this step makes sense and then explain this in your notebook. Inline explanations in the notebook are required — your reasoning must be visible.

10. Submission Instructions

Your notebook should be submitted through the Moodle platform. You can make several submissions, taking into account that a submission replaces the previous one.
If the data can be easily retrieved from the provided address, and was not subject to change during the assignment, just provide the link to the data in your assignment notebook. Otherwise, submit the data together with your work.

11. Policies

Students may share and/or exchange ideas with each other about the assignments and/or solving them. However, the work turned in must correspond to the individual effort of each group. The following situations are considered fraud:

- Partially copied work
- Facilitating copying by sharing files
- Using other people's material without mentioning the source.

In case of detection of any type of fraud, the work in question will not be evaluated, and will be sent to the Pedagogical Commission or the Pedagogical Council, depending on the gravity of the situation, which will decide the sanction to be applied to the students involved. Plagiarism tools will be used for automatic copy detection.